Back **Bias, Variance and Model Complexity**
Practice Assignment • 10 min

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Module 6 Introduction

Lesson 1: Bias, Variance and Model Complexity

- **Reading:** Bias, Variance and Model Complexity Readings
  1h 15m
- **Video:** Bias, Variance and Model Complexity
  10 min
- **Video:** The Bias-Variance Decomposition
  8 min
- **Practice Assignment:** Bias, Variance and Model Complexity
  Submitted
- **Practice Assignment:** Bias, Variance and Model Complexity Quiz 2
  1h
- **Lab:** Coding Example
  1h

Lesson 2: Bayesian Approach and BIC

Lesson 3: Vapnik-Chernovenkis (VC) Dimension

Lesson 4: Cross Validation

Lesson 5: Bootstrapping

Module 6 Summative Assessment

Module 6 Summary

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

Instructions

1. What does the term "bias" refer to in the context of the bias-variance tradeoff? **1 / 1 point**

○ The rate at which a model learns during training.
● The error due to overly simplistic assumptions in the model.
○ The degree of overfitting in a model.
○ The error due to random fluctuations in the training data.

Correct
Correct! Bias is the error introduced by approximating a real-world problem by a too-simple model.

2. What does "optimism of the training error rate" refer to in model evaluation? **1 / 1 point**

○ The error rate of a model on test data.
○ The accuracy rate a model achieves on training data.
○ The overestimation of a model's error rate during training.
● The tendency of the training error rate to underestimate the true error rate on new data.

Correct
Correct! Optimism refers to the lower error rate observed on the training data compared to what is expected on new, unseen data.

3. Why is it important to account for the optimism of the training error rate? **1 / 1 point**

○ To reduce the computational complexity of the model.
○ To increase the training error rate deliberately.
● To obtain a more accurate estimate of how the model will perform on unseen data.
○ To ensure the model is perfectly accurate on training data.

Correct
Correct! Accounting for optimism helps in estimating the true performance of the model on new data.

4. What typically happens to the optimism in the training error rate as model complexity increases? **1 / 1 point**

○ It becomes zero, indicating no difference between training and test error rates.
● It tends to increase, leading to a larger gap between training and test error rates.
○ It remains constant regardless of model complexity.
○ It decreases, making the training error rate more reliable.

Correct
Correct! Higher complexity models often have more optimistic training error rates.

5. In the context of model selection, how can optimism of the training error rate be addressed? **1 / 1 point**

○ By using the same dataset for both training and testing.
● Through techniques like cross-validation or regularization.
○ By training the model on as much data as possible.
○ By always choosing the simplest possible model.

Correct
Correct! These techniques help to mitigate the optimism of the training error rate.

6. What is the impact of high optimism on the training error rate when deploying a model in a real-world scenario? **1 / 1 point**

● It can lead to overestimating the model's performance, resulting in poorer results than expected.
○ It guarantees the model's success in new environments.
○ It ensures the model is robust to changes in data.
○ It decreases the overall error rate of the model in practice.

Correct
Correct! High optimism can cause overconfidence in the model's real-world performance.

Bias, Variance and Model Complexity

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

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Nov 16, 2:21 PM IST	Unlimited

Try again

Your grade
To pass you need at least 80%. We keep your highest score.

100%

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